

STUDY UNIT 9

Logic: Truth Tables

Complete study notes - Thorough coverage of Unit 9 with definitions, truth tables, logical laws, translation rules, full worked examples, official-style self-assessment solutions, exam traps, and a revision checklist. Every major topic/category includes an added example.

Coverage: declarative statements; propositions; simple and compound statements; connectives; conjunction; disjunction; negation; biconditional; implication; English-to-symbolic translation; truth-table construction; tautologies; contradictions; logical equivalence; equivalence laws; De Morgan's laws; implication and biconditional rewriting; exclusive OR; symbolic simplification; equivalence relations; worked assessment-style problems.

Prepared as expanded revision notes for COS1501 Study Unit 9.

Contents and learning outcomes

By the end of Unit 9 you should be able to do all of the following without guessing:

- Recognise declarative statements and distinguish them from questions, commands and exclamations.
- Distinguish simple (atomic) statements from compound statements and assign truth values where appropriate.
- Use and interpret the five main logical connectives: $\wedge$, $\vee$, $\neg$, $\leftrightarrow$ and $\rightarrow$.
- Translate ordinary English statements into symbolic logic, especially “if”, “only if”, “if and only if”, “neither...nor”, and “not both”.
- Construct complete truth tables with 1, 2, 3 or more basic propositions using the 2^n row rule.
- Determine whether a compound statement is a tautology, a contradiction, or neither.
- Determine and prove logical equivalence using truth tables and standard logical laws.
- Use commutative, associative, distributive, idempotent, double-negation, De Morgan, identity, domination and complement laws.
- Rewrite implication and biconditional expressions using only $\neg$, $\wedge$ and $\vee$.
- Work with exclusive OR and prove that logical equivalence is an equivalence relation.
- Simplify complex expressions and solve COS1501-style self-assessment questions.

Part	Coverage
1	Statements and truth values
2	Logical connectives
3	Conditional language and translation
4	Constructing truth tables
5	Tautology, contradiction and contingency
6	Logical equivalence and laws
7	Implication, biconditional and XOR
8	Symbolic simplification
9	Worked COS1501-style exercises
10	Revision checklist and practice

1. Declarative statements, propositions and truth values

1.1 What is a declarative statement?

A declarative statement (also called a proposition) is a sentence that conveys information. In propositional logic, we normally restrict attention to declarative statements that can be classified as either true (T) or false (F).

Questions, commands and exclamations are not declarative statements because they do not assert a fact that can be assigned a truth value.

Example - declarative versus non-declarative

“11 is a prime number” is a declarative statement and is true. “Is 11 prime?” is a question, so it is not a declarative statement. “Write down 11!” is a command, so it is not a declarative statement.

1.2 Truth values

A usable proposition has one truth value: T or F. The truth value belongs to the complete statement, not to the symbol used to name it.

$$p = \text{“8 is even”} \rightarrow p = T$$
$$q = \text{“8 is prime”} \rightarrow q = F$$

Example - assigning truth values

Let p be “South Africa is in Africa” and q be “9 is an even integer”. Then $p = T$ and $q = F$.

1.3 Not every declarative sentence is usable

Some sentences are grammatically declarative but do not have a consistent truth value. The classic self-referential sentence “This sentence is false” creates a paradox: assuming it is true makes it false, while assuming it is false makes it true. Propositional logic therefore works with statements that have a definite truth value.

Example - statement that depends on missing information

The sentence “ $x + 2 = 7$ ” is not a complete proposition until x has been fixed or quantified. If $x = 5$ it is true; if $x = 3$ it is false. Unit 10 later formalises this kind of expression using predicates and quantifiers.

1.4 Simple (atomic) and compound statements

A simple/atomic statement conveys one basic fact. A compound statement is created by combining one or more propositions with logical connectives.

$$p: \text{“It is raining.”}$$
$$q: \text{“It is cold.”}$$
$$p \wedge q: \text{“It is raining and it is cold.”}$$

Example - simple versus compound

“The server is online” is simple. “The server is online and the database is reachable” is compound because it combines two basic claims with AND.

2. Logical connectives

A logical connective is an operator used to form a compound statement from simpler statements. Unit 9 uses the following five main connectives.

Symbol	Meaning
$p \wedge q$	Conjunction: p AND q
$p \vee q$	Disjunction: p OR q (inclusive OR)
$\neg p$	Negation: NOT p
$p \leftrightarrow q$	Biconditional: p IF AND ONLY IF q (p iff q)
$p \rightarrow q$	Conditional/implication: IF p , THEN q

Example - combining connectives

Let p = "I submit the assignment" and q = "I receive a mark". The expression $p \rightarrow q$ reads: "If I submit the assignment, then I receive a mark."

2.1 Conjunction (AND)

The conjunction $p \wedge q$ is true only when both p and q are true. If either part is false, the conjunction is false.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Example - conjunction

p = "A file exists" and q = "The file is readable". If the file exists (T) but is not readable (F), then $p \wedge q = F$. Both conditions must hold.

Memory rule - AND = BOTH.

2.2 Disjunction (OR)

The disjunction $p \vee q$ is inclusive OR. It is true if at least one of p or q is true, including the case where both are true. It is false only when both are false.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Example - inclusive OR

A login is allowed if the user has an admin account or a valid access token. If both are present, the OR statement is still true.

Memory rule - OR = AT LEAST ONE. Do not confuse ordinary OR with exclusive OR.

2.3 Negation (NOT)

The negation $\neg p$ reverses the truth value of p .

p	$\neg p$
T	F
F	T

Example - negation

If p = "The printer is connected" is false, then $\neg p$ = "The printer is not connected" is true.

Memory rule - Negation flips T and F.

2.4 Biconditional (IF AND ONLY IF)

The biconditional $p \leftrightarrow q$ is true when p and q have the same truth value, and false when they have different truth values. "If and only if" is often shortened to iff.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Example - biconditional

Let p = "n is even" and q = "n is divisible by 2", for an integer n . These conditions always agree, so $p \leftrightarrow q$ is true for every integer n .

Memory rule - Biconditional = SAME. Same truth values give T; different truth values give F.

2.5 Conditional / implication

The conditional $p \rightarrow q$ reads "If p , then q ". The statement p is the antecedent/hypothesis and q is the consequent/conclusion.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

The conditional is false in exactly one case: p is true and q is false. When p is false, the conditional is considered true because the promised condition has not been violated.

Example - implication

"If a number is divisible by 4, then it is even." For $n = 12$, $T \rightarrow T = T$. For a hypothetical case where something is divisible by 4 but not even, $T \rightarrow F$ would make the implication false. If n is not divisible by 4, the statement is not violated, so the implication row is true.

Memory rule - Implication is false only for $T \rightarrow F$.

2.6 Antecedent and consequent

$$p \rightarrow q$$

p is the antecedent (hypothesis). q is the consequent (conclusion). The direction matters.

Example - identifying the parts

In "If the password is correct, then access is granted", p = "the password is correct" is the antecedent and q = "access is granted" is the consequent.

3. Translating English into symbolic logic

3.1 “If”, “only if” and “if and only if”

This is one of the most important language traps in Unit 9.

English form	Symbolic form
“p if q”	$q \rightarrow p$
“p only if q”	$p \rightarrow q$
“p if and only if q”	$p \leftrightarrow q$

A useful way to remember “only if” is that the condition after only if becomes necessary for the first statement.

Example - if versus only if

Let p = “I enter the lab” and q = “I have an access card”. “I enter the lab if I have an access card” becomes $q \rightarrow p$. “I enter the lab only if I have an access card” becomes $p \rightarrow q$.

3.2 Translating AND, OR and NOT

Translate the English structure before trying to calculate truth values. Parentheses matter because they show which part is being negated or combined.

Example - basic translation

Let p = “It is sunny” and q = “It is humid”. Then “It is sunny and not humid” is $p \wedge \neg q$; “It is humid or sunny” is $q \vee p$; “It is false that it is humid” is $\neg q$.

3.3 “Neither...nor”

“Neither p nor q ” means that p is false and q is false.

$$\neg p \wedge \neg q$$

By De Morgan’s law this is equivalent to $\neg(p \vee q)$.

Example - neither...nor

“The device is neither charging nor connected” becomes $\neg p \wedge \neg q$, where p = “the device is charging” and q = “the device is connected”.

3.4 “Not both”

“Not both p and q ” means it is not the case that p and q are simultaneously true.

$$\neg(p \wedge q)$$

This is not the same as “neither p nor q ”. It allows one of the two statements to be true.

Example - not both

“Not both servers are offline” means $\neg(p \wedge q)$. It permits server A offline/server B online, A online/B offline, or both online. It excludes only the case where both are offline.

3.5 A full translation set

Let p = “It is sunny” and q = “It is humid”.

Statement	Symbolic form
It is sunny and it is not humid.	$p \wedge \neg q$
It is humid or it is sunny.	$q \vee p$
It is false that it is humid.	$\neg q$
It is false that it is sunny and humid.	$\neg(p \wedge q)$
It is neither sunny nor humid.	$\neg p \wedge \neg q$
It is not the case that if it is sunny then it is humid.	$\neg(p \rightarrow q)$
It is humid if it is sunny.	$p \rightarrow q$
It is humid only if it is sunny.	$q \rightarrow p$
It is sunny if and only if it is humid.	$p \leftrightarrow q$
If neither sunny nor humid, then it is not sunny.	$\neg(p \vee q) \rightarrow \neg p$

Example - translation with brackets

“If it is not true that the network is online or the backup is available, then the network is not online” becomes $\neg(p \vee q) \rightarrow \neg p$.

4. Constructing truth tables

4.1 Number of rows: the 2^n rule

If a compound statement contains n distinct simple propositions, a complete truth table has:

$$2^n \text{ rows}$$

This is because each proposition has two possible truth values, and the table must list every combination.

Number of variables	Rows
1 proposition	$2^1 = 2$ rows
2 propositions	$2^2 = 4$ rows
3 propositions	$2^3 = 8$ rows
4 propositions	$2^4 = 16$ rows

Example - row count

The expression $(p \rightarrow q) \vee r$ contains three distinct propositions p , q and r . Therefore its truth table needs $2^3 = 8$ rows.

4.2 Standard truth-value pattern

For three propositions, the standard pattern is: first column changes halfway; second changes every two rows; third alternates every row.

p	q	
T	T	T
T	T	F
T	F	T
T	F	F

p	q	
F	T	T
F	T	F
F	F	T
F	F	F

Example - pattern check

For four variables there are 16 rows. The p column has 8 T then 8 F; q has 4 T, 4 F repeated; r has 2 T, 2 F repeated; s alternates T, F.

4.3 Intermediate columns

Never try to evaluate a complicated statement all at once. Create a column for every meaningful subexpression. Work from the innermost operations outward.

For $p \wedge \neg q$, use columns p, q, $\neg q$, then $p \wedge \neg q$.

p	q	$\neg q$	$p \wedge \neg q$
T	T	F	F
T	F	T	T
F	T	F	F
F	F	T	F

Example - intermediate columns

For $\neg p \rightarrow (q \wedge r)$, calculate $\neg p$ first, calculate $q \wedge r$ second, and only then calculate the implication.

4.4 Order of evaluation

Parentheses determine the exact order. A practical workflow is:

- List the basic propositions.
- Calculate negations attached directly to propositions.
- Evaluate expressions inside the deepest parentheses.
- Move outward one connective at a time.
- Use the final column to answer the actual question.

Example - order of work

For $[\neg p \rightarrow (q \wedge r)] \vee r$: first find $\neg p$; then $q \wedge r$; then $\neg p \rightarrow (q \wedge r)$; finally OR that result with r.

4.5 Complete two-variable worked table

Construct the truth table for $(p \vee q) \wedge \neg p$.

p	q	$p \vee q$	$\neg p$	$(p \vee q) \wedge \neg p$
T	T	T	F	F
T	F	T	F	F
F	T	T	T	T
F	F	F	T	F

Example conclusion

The final column contains both T and F. Therefore the statement is neither a tautology nor a contradiction.

4.6 Complete three-variable worked table

Construct the truth table for $[\neg p \rightarrow (q \wedge r)] \vee r$.

p	q	r	$\neg p$	$q \wedge r$	$\neg p \rightarrow (q \wedge r)$	Final
T	T	T	F	T	T	T
T	T	F	F	F	T	T
T	F	T	F	F	T	T
T	F	F	F	F	T	T
F	T	T	T	T	T	T
F	T	F	T	F	F	F
F	F	T	T	F	F	T
F	F	F	T	F	F	F

Example conclusion

Because the final column contains both truth values, the expression is a contingency (neither tautology nor contradiction).

5. Relationships between statements

5.1 Tautology

A tautology is a compound statement that is true for every possible assignment of truth values.

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

Example - tautology

The law of excluded middle $p \vee \neg p$ is always true: p must either be true or false, so one side of the OR is always true.

5.2 Contradiction

A contradiction is a compound statement that is false for every possible assignment of truth values.

p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

Example - contradiction

“The light is on and the light is not on” has the form $p \wedge \neg p$ and is always false.

5.3 Neither: contingency

If the final truth-table column contains a mixture of T and F, the statement is neither a tautology nor a contradiction. It is often called a contingency.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Example - contingency

$p \wedge q$ is true in one row and false in three rows, so it is neither always true nor always false.

Fast exam test - All T $\rightarrow$ tautology. All F $\rightarrow$ contradiction. Mixed T/F $\rightarrow$ neither.

5.4 Logical equivalence

Two statements A and B are logically equivalent, written $A \equiv B$, when they have exactly the same truth value for every assignment of their basic propositions. Equivalently, $A \leftrightarrow B$ must be a tautology.

A single truth table is usually the fastest proof: compute A and B side by side and compare their final columns.

p	q	$p \vee q$	$q \vee p$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	F

Example - logical equivalence

The last two columns are identical, therefore $p \vee q \equiv q \vee p$. This is the commutative law for OR.

5.5 Biconditional versus logical equivalence

Do not confuse the symbols. $p \leftrightarrow q$ is itself a compound statement with a truth value on each row. $A \equiv B$ is a claim that two expressions always have the same truth behaviour.

Example - distinction

For one specific row $p = T$ and $q = T$, $p \leftrightarrow q$ is true. That does not by itself prove $p \equiv q$. To prove equivalence you must show agreement on every row.

6. Laws of logical equivalence

These laws allow you to manipulate compound statements algebraically. You should know the forms and be able to name the law used at each step.

6.1 Commutative laws

$$p \vee q \equiv q \vee p$$

$$p \wedge q \equiv q \wedge p$$

The order of operands can be swapped for AND and OR.

Example - commutative

$(a \wedge b) \vee c$ can be rewritten as $(b \wedge a) \vee c$ because $a \wedge b \equiv b \wedge a$.

6.2 Associative laws

$$p \vee (q \vee r) \equiv (p \vee q) \vee r$$

$$p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$$

When the same connective is repeated, grouping can change.

Example - associative

$p \wedge (q \wedge r)$ can be regrouped as $(p \wedge q) \wedge r$ without changing its truth value.

6.3 Distributive laws

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

Example - distributive

$p \wedge (q \vee r)$ expands to $(p \wedge q) \vee (p \wedge r)$, just as multiplication distributes over addition in ordinary algebra.

6.4 Idempotent laws

$$p \vee p \equiv p$$

$$p \wedge p \equiv p$$

Example - idempotent

“The file exists OR the file exists” adds no new information, so $p \vee p$ simplifies to p .

6.5 Double negation

$$\neg(\neg p) \equiv p$$

Example - double negation

$\neg(\neg q)$ simplifies directly to q . Two logical NOTs cancel.

6.6 De Morgan's laws

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

When a negation moves through parentheses, negate each component and switch $\vee \leftrightarrow \wedge$.

Example - De Morgan

$$\neg(p \vee \neg q) \equiv \neg p \wedge \neg(\neg q) \equiv \neg p \wedge q.$$

Memory rule - Push NOT inside, flip the connective, and negate each part.

6.7 Complement / inverse laws

$$p \vee \neg p \equiv T$$

$$p \wedge \neg p \equiv F$$

Example - complement

$x \wedge \neg x$ is always false, so any expression containing it as a mandatory AND condition may simplify further.

6.8 Identity laws

$$p \vee F \equiv p$$

$$p \wedge T \equiv p$$

Example - identity

$(q \wedge T) \vee F$ simplifies first to $q \vee F$, then to q .

6.9 Domination laws

$$p \vee T \equiv T$$

$$p \wedge F \equiv F$$

Example - domination

$p \vee T$ is true regardless of p , so it simplifies to T .

6.10 Negation of constants

$$\neg T \equiv F$$

$$\neg F \equiv T$$

Example - constants

$\neg(p \vee \neg p) = \neg T = F$, so the negation of a tautology is a contradiction.

6.11 Master law table

Law	Equivalence
Commutative	$p \vee q \equiv q \vee p$; $p \wedge q \equiv q \wedge p$
Associative	$p \vee (q \vee r) \equiv (p \vee q) \vee r$; $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$
Distributive	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$; $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
Idempotent	$p \vee p \equiv p$; $p \wedge p \equiv p$
Double negation	$\neg \neg p \equiv p$
De Morgan	$\neg(p \vee q) \equiv \neg p \wedge \neg q$; $\neg(p \wedge q) \equiv \neg p \vee \neg q$
Complement	$p \vee \neg p \equiv T$; $p \wedge \neg p \equiv F$
Identity	$p \vee F \equiv p$; $p \wedge T \equiv p$
Domination	$p \vee T \equiv T$; $p \wedge F \equiv F$

Example - chained simplification

$(p \wedge T) \vee (q \wedge F) \equiv p \vee F \equiv p$. Identity and domination laws remove the constants.

7. Rewriting implication, biconditional and exclusive OR

7.1 Implication using OR and NOT

$$p \rightarrow q \equiv \neg p \vee q$$

This equivalence is fundamental. It lets us eliminate the $\rightarrow$ connective.

p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Example - implication rewrite

$(r \rightarrow s) \wedge r$ can be rewritten as $(\neg r \vee s) \wedge r$.

7.2 Negating an implication

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

Derivation: $\neg(p \rightarrow q) \equiv \neg(\neg p \vee q) \equiv \neg\neg p \wedge \neg q \equiv p \wedge \neg q$.

Example - negated implication

“It is not true that if the program runs then the test passes” becomes $p \wedge \neg q$: the program runs, but the test does not pass.

7.3 Biconditional as two implications

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

A biconditional requires both directions to hold.

p	q	$p \rightarrow q$	$q \rightarrow p$	Both	$p \leftrightarrow q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

Example - biconditional rewrite

“n is even iff n is divisible by 2” means both “if n is even, then n is divisible by 2” and “if n is divisible by 2, then n is even”.

7.4 Biconditional using only $\neg$, $\wedge$ and $\vee$

Replace both implications in the previous identity:

$$p \leftrightarrow q \equiv (\neg p \vee q) \wedge (\neg q \vee p)$$

Example - remove $\leftrightarrow$ and $\rightarrow$

$a \leftrightarrow b$ can be written as $(\neg a \vee b) \wedge (\neg b \vee a)$, which uses only NOT, OR and AND.

7.5 Why $\neg$, $\wedge$ and $\vee$ are enough

Because $\rightarrow$ can be replaced by $\neg p \vee q$ and $\leftrightarrow$ can be replaced by two implications, every expression in this unit can be rewritten using only the basic connectives $\neg$, $\wedge$ and $\vee$.

Example - eliminate connectives

$(p \leftrightarrow q) \rightarrow r$ can first become $\neg[(\neg p \vee q) \wedge (\neg q \vee p)] \vee r$, which contains no $\leftrightarrow$ or $\rightarrow$.

7.6 Exclusive OR (XOR)

Exclusive OR means “p or q, but not both”. In the COS1501 material it is denoted by $p + q$.

p	q	$p + q$
T	T	F
T	F	T
F	T	T
F	F	F

$$p + q \equiv (p \vee q) \wedge \neg(p \wedge q)$$

Example - XOR

A switch is in exactly one of two modes. If mode A is on and mode B is off, XOR is true; if both modes are on, XOR is false.

Compare - Ordinary OR is true for T,T. Exclusive OR is false for T,T.

8. Logical equivalence as a relation and symbolic simplification

8.1 Reflexive property

Logical equivalence is reflexive because every statement has the same truth values as itself.

$$p \equiv p$$

Example - reflexive

The truth table for p and p is identical row for row, so $p \leftrightarrow p$ is always true.

8.2 Symmetric property

If $p \equiv q$, then $q \equiv p$. Reversing the order does not change the fact that the truth tables are identical.

$$p \equiv q \Rightarrow q \equiv p$$

Example - symmetric

If $p \vee q \equiv q \vee p$, then equally $q \vee p \equiv p \vee q$.

8.3 Transitive property

If $p \equiv q$ and $q \equiv r$, then $p \equiv r$.

$$(p \equiv q) \wedge (q \equiv r) \Rightarrow p \equiv r$$

Example - transitive

If $A \equiv \neg\neg A$ and $\neg\neg A \equiv B$, then $A \equiv B$.

Therefore logical equivalence $\equiv$ is an equivalence relation: reflexive, symmetric and transitive.

8.4 Symbolic simplification workflow

- Remove implications if doing so exposes De Morgan or complement patterns.
- Push negations inward using De Morgan's laws.
- Cancel double negations.
- Use complement laws $p \vee \neg p = T$ and $p \wedge \neg p = F$.
- Apply identity/domination laws to remove T and F.
- Use associative/commutative/distributive laws to regroup when needed.

Example - simplification

$\neg(p \vee \neg q) \equiv \neg p \wedge \neg\neg q \equiv \neg p \wedge q$. First use De Morgan; then double negation.

8.5 More advanced simplification

Simplify $\neg(p \rightarrow q)$.

$$\begin{aligned}\neg(p \rightarrow q) \\ &\equiv \neg(\neg p \vee q) \text{ [implication]} \\ &\equiv \neg\neg p \wedge \neg q \text{ [De Morgan]} \\ &\equiv p \wedge \neg q \text{ [double negation]}\end{aligned}$$

Example - interpretation

The only way "if p then q" can fail is for p to be true while q is false. The simplified form $p \wedge \neg q$ expresses exactly that.

8.6 Contrapositive equivalence

An implication is logically equivalent to its contrapositive:

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

p	q	$p \rightarrow q$	$\neg q$	$\neg p$	$\neg q \rightarrow \neg p$
T	T	T	F	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Example - contrapositive

"If n is divisible by 4, then n is even" has contrapositive "If n is not even, then n is not divisible by 4." These two statements are logically equivalent.

9. Worked COS1501-style self-assessment

This section works through the major exercise types that appear in Unit 9. The wording is kept concise, but every step needed for assessment is shown.

9.1 Translation exercise

Let p = "It is sunny" and q = "It is humid".

English idea	Answer
(a) Sunny and not humid	$p \wedge \neg q$
(b) Humid or sunny	$q \vee p$
(c) False that humid	$\neg q$
(d) False that sunny and humid	$\neg(p \wedge q)$
(e) Neither sunny nor humid	$\neg p \wedge \neg q$
(f) Not the case that if sunny then humid	$\neg(p \rightarrow q)$
(g) Humid if sunny	$p \rightarrow q$
(h) Humid only if sunny	$q \rightarrow p$
(i) Sunny iff humid	$p \leftrightarrow q$
(j) If neither sunny nor humid, then not sunny	$\neg(p \vee q) \rightarrow \neg p$

Added example

Let a = "the user is authenticated" and b = "the user has access". "The user has access only if authenticated" becomes $b \rightarrow a$.

9.2 Truth table: $[(\neg q) \rightarrow (\neg p)] \rightarrow (p \rightarrow q)$

p	q	$\neg q$	$\neg p$	$\neg q \rightarrow \neg p$	$p \rightarrow q$	Final
T	T	F	F	T	T	T
T	F	T	F	F	F	T
F	T	F	T	T	T	T
F	F	T	T	T	T	T

All final values are T, so the expression is a tautology.

Added example

This is another way of seeing the contrapositive connection: $\neg q \rightarrow \neg p$ and $p \rightarrow q$ always have matching truth behaviour.

9.3 Truth table: $[\neg p \rightarrow (q \wedge \neg q)] \rightarrow p$

p	q	$\neg p$	$\neg q$	$q \wedge \neg q$	$\neg p \rightarrow (\dots)$	Final
T	T	F	F	F	T	T
T	F	F	T	F	T	T
F	T	T	F	F	F	T
F	F	T	T	F	F	T

All final values are T, so this statement is a tautology. Notice $q \wedge \neg q$ is always F.

Added example

Whenever an inner subexpression is a contradiction, simplifying it first can make a long truth table much easier.

9.4 Truth table: $p \vee \neg p$

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

This is a tautology.

Added example

$q \vee \neg q$ is the same pattern and is also a tautology.

9.5 Truth table: $[p \wedge (p \rightarrow q)] \rightarrow q$

p	q	$p \rightarrow q$	$p \wedge (p \rightarrow q)$	Final
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

All final values are T. This truth-table form reflects the valid reasoning pattern commonly called modus ponens.

Added example

If “I study” is true and “if I study then I pass” is true, the conclusion “I pass” must be true.

9.6 Truth table: $(p \vee q) \wedge (\neg p \vee \neg q)$

p	q	$\neg p$	$\neg q$	$p \vee q$	$\neg p \vee \neg q$	Final
T	T	F	F	T	F	F
T	F	F	T	T	T	T
F	T	T	F	T	T	T
F	F	T	T	F	T	F

The final pattern F,T,T,F is exactly exclusive OR: one is true, but not both.

Added example

This expression gives a way to build XOR from ordinary OR, AND and NOT.

9.7 Three-variable truth table: $(\neg p \rightarrow [q \wedge r]) \vee r$

p	q	r	$\neg p$	$q \wedge r$	$\neg p \rightarrow (q \wedge r)$	Final
T	T	T	F	T	T	T
T	T	F	F	F	T	T
T	F	T	F	F	T	T
T	F	F	F	F	T	T
F	T	T	T	T	T	T
F	T	F	T	F	F	F

p	q	r	$\neg p$	$q \wedge r$	$\neg p \rightarrow (q \wedge r)$	Final
F	F	T	T	F	F	T
F	F	F	T	F	F	F

The final column is mixed, so the expression is neither a tautology nor a contradiction.

Added example

In a three-variable table, verify the base columns first. A single incorrect T/F pattern can corrupt every later column.

9.8 Three-variable biconditional comparison

Evaluate $(p \rightarrow [q \wedge r]) \leftrightarrow ([p \rightarrow q] \vee [p \rightarrow r])$.

p	q	r	$q \wedge r$	$p \rightarrow q$	$p \rightarrow r$	$p \rightarrow (q \wedge r)$	OR side	Final $\leftrightarrow$
T	T	T	T	T	T	T	T	T
T	T	F	F	T	F	F	T	F
T	F	T	F	F	T	F	T	F
T	F	F	F	F	F	F	F	T
F	T	T	T	T	T	T	T	T
F	T	F	F	T	T	T	T	T
F	F	T	F	T	T	T	T	T
F	F	F	F	T	T	T	T	T

The final biconditional is false on two rows, so the two sides are not logically equivalent.

Added example

To test equivalence of large expressions, compare the two side columns directly before calculating a biconditional. One differing row is enough to disprove equivalence.

9.9 Rewrite $p \leftrightarrow q$ using only $\neg$, $\wedge$ and $\vee$

$$\begin{aligned}
 p &\leftrightarrow q \\
 &\equiv (p \rightarrow q) \wedge (q \rightarrow p) \\
 &\equiv (\neg p \vee q) \wedge (\neg q \vee p)
 \end{aligned}$$

Added example

$$r \leftrightarrow s \equiv (\neg r \vee s) \wedge (\neg s \vee r).$$

9.10 Show that $\equiv$ is an equivalence relation

Reflexive: $p \equiv p$. Symmetric: if $p \equiv q$, then $q \equiv p$. Transitive: if $p \equiv q$ and $q \equiv r$, then $p \equiv r$. Therefore $\equiv$ is an equivalence relation.

Added example

If $A \equiv B$ and $B \equiv C$, every truth-value assignment gives A and B the same value and B and C the same value, so A and C must also match.

9.11 Exclusive disjunction

p	q	$p + q$
T	T	F
T	F	T
F	T	T
F	F	F

Added example

If exactly one of two alarms is active, $p + q$ is true. If neither or both are active, it is false.

9.12 Find an equivalent statement to $\neg(p \vee \neg q)$

$$\begin{aligned}
 &\neg(p \vee \neg q) \\
 &\equiv \neg p \wedge \neg(\neg q) \text{ [De Morgan]} \\
 &\equiv \neg p \wedge q \text{ [double negation]}
 \end{aligned}$$

Added example

Similarly, $\neg(\neg p \vee q) \equiv p \wedge \neg q$.

9.13 Push negations inside parentheses - example A

Rewrite $\neg[(p \vee q \vee \neg q) \wedge (q \wedge \neg p)]$ so that no $\neg$ remains outside a compound parenthesis.

$$\begin{aligned}
 &\neg[(p \vee q \vee \neg q) \wedge (q \wedge \neg p)] \\
 &\equiv \neg(p \vee q \vee \neg q) \vee \neg(q \wedge \neg p) \\
 &\equiv (\neg p \wedge \neg q \wedge \neg\neg q) \vee (\neg q \vee \neg\neg p) \\
 &\equiv (\neg p \wedge \neg q \wedge q) \vee (\neg q \vee p)
 \end{aligned}$$

Added example

At each De Morgan step, both things happen: every component gets negated and the connective flips.

9.14 Push negations inside parentheses - example B

Rewrite $\neg[(p \vee (p \rightarrow q)) \vee (p \wedge q)]$.

$$\begin{aligned}
 &\neg[(p \vee (p \rightarrow q)) \vee (p \wedge q)] \\
 &\equiv \neg[(p \vee (\neg p \vee q)) \vee (p \wedge q)] \text{ [implication]} \\
 &\equiv \neg[p \vee (\neg p \vee q)] \wedge \neg(p \wedge q) \text{ [De Morgan]} \\
 &\equiv [\neg p \wedge \neg(\neg p \vee q)] \wedge (\neg p \vee \neg q) \\
 &\equiv [\neg p \wedge (p \wedge \neg q)] \wedge (\neg p \vee \neg q)
 \end{aligned}$$

The expression contains both p and $\neg p$ as required AND conditions, so it is in fact contradictory and can ultimately simplify to F.

Added example

Recognising $p \wedge \neg p$ after a few transformations can collapse a large expression immediately to F.

9.15 Determine whether two statements are equivalent

Compare $A = \neg p \wedge (\neg p \wedge \neg q)$ with $B = \neg(p \vee (p \rightarrow q))$.

Simplify A:

$$\neg p \wedge (\neg p \wedge \neg q) \equiv \neg p \wedge \neg q$$

Simplify B:

$$\begin{aligned}\neg(p \vee (p \rightarrow q)) &\equiv \neg(p \vee (\neg p \vee q)) \\ &\equiv \neg(p \vee \neg p \vee q) \equiv \neg(T \vee q) \equiv \neg T \equiv F\end{aligned}$$

A is not always false, while B is always false. Therefore the two statements are not equivalent.

Added example

A single counter-row is enough to disprove equivalence. For $p = F$ and $q = F$, A is T but B is F.

10. Exam revision and completion checklist

10.1 One-page truth-table memory sheet

p	q	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$	$p + q$
T	T	T	T	T	T	F
T	F	F	T	F	F	T
F	T	F	T	T	F	T
F	F	F	F	T	T	F

Negation: T becomes F and F becomes T.

10.2 Core formulas to memorise

- Rows in a truth table: 2^n
- $p \rightarrow q \equiv \neg p \vee q$
- $\neg(p \rightarrow q) \equiv p \wedge \neg q$
- $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
- $p \leftrightarrow q \equiv (\neg p \vee q) \wedge (\neg q \vee p)$
- $\neg(p \vee q) \equiv \neg p \wedge \neg q$
- $\neg(p \wedge q) \equiv \neg p \vee \neg q$
- $p \rightarrow q \equiv \neg q \rightarrow \neg p$
- $p + q \equiv (p \vee q) \wedge \neg(p \wedge q)$

Example - using the memory sheet

To evaluate $T \rightarrow F$, use the implication rule: this is the one false implication case. To evaluate $F \leftrightarrow F$, use the biconditional rule: same values means true.

10.3 Common exam traps

Trap	Correct rule
Ordinary OR	$T \vee T = T$. OR is inclusive unless "exclusive OR" is stated.
Implication	Do not mark $F \rightarrow F$ as false. Only $T \rightarrow F$ is false.
Only if	"p only if q" is $p \rightarrow q$, not $q \rightarrow p$.
Negation scope	$\neg(p \wedge q)$ is not $\neg p \wedge \neg q$. De Morgan changes AND to OR.

Trap	Correct rule
Equivalence	One matching row does not prove equivalence; all rows must match.
Truth-table rows	Count distinct simple propositions, not the number of symbols.
Brackets	Never ignore parentheses. They decide which subexpression is evaluated first.

Example - trap check

"It is not the case that both p and q" is $\neg(p \wedge q)$, which is equivalent to $\neg p \vee \neg q$, not $\neg p \wedge \neg q$.

10.4 Completion checklist

- ✓ Define a declarative statement/proposition.
- ✓ Identify non-declarative sentences.
- ✓ Assign T or F to a usable proposition.
- ✓ Distinguish atomic and compound statements.
- ✓ Know all five main connectives and their symbols.
- ✓ Know the truth table for AND.
- ✓ Know the truth table for inclusive OR.
- ✓ Know negation.
- ✓ Know the biconditional truth table.
- ✓ Know the implication truth table.
- ✓ Identify antecedent and consequent.
- ✓ Translate "if", "only if" and "iff".
- ✓ Translate "neither...nor" and "not both".
- ✓ Use 2^n to find truth-table rows.
- ✓ Build correct T/F base-column patterns.
- ✓ Use intermediate columns.
- ✓ Classify tautology, contradiction or neither.
- ✓ Prove logical equivalence using identical columns.
- ✓ Use all standard equivalence laws.
- ✓ Apply both De Morgan laws.
- ✓ Rewrite $p \rightarrow q$ as $\neg p \vee q$.
- ✓ Negate an implication as $p \wedge \neg q$.
- ✓ Rewrite $p \leftrightarrow q$ using two implications.
- ✓ Rewrite $\leftrightarrow$ using only $\neg$, $\wedge$ and $\vee$.
- ✓ Construct and interpret XOR.
- ✓ Explain why $\equiv$ is reflexive, symmetric and transitive.
- ✓ Simplify expressions step by step and name the law used.

Example - readiness test

If you can take an unfamiliar three-variable expression, construct its 8-row truth table correctly, and then justify whether it is a tautology/contradiction/neither, you have mastered the central procedural skill of Unit 9.

10.5 Practice questions

Try these without looking at the solutions immediately.

- 1. Let p = “the app is open” and q = “the user is logged in”. Translate: “The app is open only if the user is logged in.”
- 2. Construct the truth table for $\neg p \vee q$.
- 3. Determine whether $(p \wedge q) \rightarrow p$ is a tautology.
- 4. Simplify $\neg(p \wedge \neg q)$.
- 5. Rewrite $p \leftrightarrow q$ using only $\neg$, $\wedge$ and $\vee$.
- 6. Determine whether $p \rightarrow q$ and $q \rightarrow p$ are logically equivalent.
- 7. Construct the XOR truth table.
- 8. How many rows are required for a truth table with p , q , r and s ?

Answers

- 1. $p \rightarrow q$.
- 2. Same truth values as $p \rightarrow q$: T,T; T,F gives F; F,T gives T; F,F gives T.
- 3. Yes. If $p \wedge q$ is true, then p is necessarily true; if $p \wedge q$ is false, the implication is true.
- 4. $\neg p \vee q$ (De Morgan + double negation).
- 5. $(\neg p \vee q) \wedge (\neg q \vee p)$.
- 6. No. For $p = T$, $q = F$, $p \rightarrow q = F$ while $q \rightarrow p = T$.
- 7. F, T, T, F for rows TT, TF, FT, FF.
- 8. $2^4 = 16$ rows.

10.6 Final Unit 9 summary

Unit 9 is the foundation of propositional logic. The key idea is that a complex logical statement can be reduced to precise truth conditions. Truth tables let you test every possible case; logical laws let you reach the same conclusions more efficiently through symbolic manipulation. You should be comfortable moving in both directions: from English to symbols, and from symbols back to their logical meaning.

Final standard - For exam readiness, you should know the five connective tables from memory, translate conditional language correctly, construct 2^n truth tables without missing rows, classify final columns, and simplify expressions using named equivalence laws.

Source basis

These notes are an expanded teaching version based on the COS1501 Study Unit 9 syllabus and study-guide topics. The explanations and added examples are written specifically for revision rather than copied from the study guide.